

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – XI
PAPER –1
TEST DATE: 23-08-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1.
Sol.

B
Let $I = kt$
Induce electric field at a distance r ($R < r < 2R$) from the axis is

$$E = \frac{R^2}{2r} \frac{dB_1}{dt} + \frac{r}{2} \frac{dB_2}{dt}$$

$$= \frac{R^2}{2r} \frac{d}{dt}(\mu_0 n 2kt) + \frac{r}{2} \frac{d}{dt}(\mu_0 n kt) = \left(\frac{R^2}{r} + \frac{r}{2} \right) \mu_0 nk$$

Velocity of the charged particle at any time t ,

$$v = \frac{qE}{m} t = \frac{q}{m} \left(\frac{R^2}{r} + \frac{r}{2} \right) \mu_0 nk$$

$$\frac{mv^2}{r} = qvB \Rightarrow \frac{mv}{r} = qB$$

$$\frac{m}{r} (\mu_0 nk) \frac{qt}{m} \left(\frac{R^2}{r} + \frac{r}{2} \right) = q \mu_0 n kt$$

$$\Rightarrow \frac{R^2}{r} = \frac{r}{2} \Rightarrow r = \sqrt{2}R$$

2.
Sol.

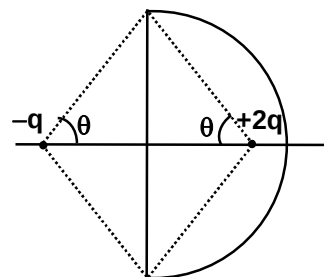
D
Flux on the plane surface $\phi_1 = \frac{3q}{2\epsilon_0} (1 - \cos \theta)$

Flux on the curved surface

$$\phi_2 = \frac{2q}{\epsilon_0} - \frac{3q}{2\epsilon_0} (1 - \cos \theta)$$

$$\therefore \phi_1 = \phi_2$$

$$\Rightarrow \frac{3q}{2\epsilon_0} (1 - \cos \theta) = \frac{2q}{\epsilon_0} - \frac{3q}{2\epsilon_0} (1 - \cos \theta)$$



$$\cos\theta = \frac{1}{3} \Rightarrow \frac{x}{\sqrt{R^2 + x^2}} = \frac{1}{3}$$

$$\Rightarrow x = \frac{R}{2\sqrt{2}}$$

3. A

 Sol. Consider a small element in the shape ring of the radius r and thickness dr of the disc. The torque on the ring due to viscous force

$$d\tau = \eta 2\pi r dr \frac{\omega r}{d}$$

Total torque on the disc due to viscous force

$$\tau = \int d\tau = \int_0^R \frac{2\pi\omega\eta}{d} r^3 dr = \frac{\pi\omega\eta R^4}{2d}$$

 External power required to rotate the disc with uniform angular velocity ω

$$P_{\text{ext}} = \tau_{\text{ext}}\omega = \frac{\pi\omega^2\eta R^4}{2d}$$

4. C

 Sol. Induced emf $\varepsilon = B\ell v = 2Bxv$

$$\text{Induced current } I = \frac{\varepsilon}{R} = \frac{2Bxv}{\lambda 2x(\sqrt{2} + 1)}$$

Magnetic force on the rod

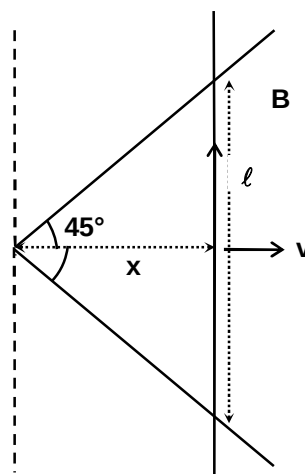
$$F = -I\ell B = \frac{-2B^2xv}{2\lambda x(\sqrt{2} + 1)} 2x$$

$$\text{Acceleration } a = \frac{F}{m} = \frac{-2B^2xv}{\lambda(\sqrt{2} + 1)m}$$

$$\Rightarrow v \frac{dv}{dx} = \frac{-2B^2xv}{\lambda(\sqrt{2} + 1)m}$$

$$\Rightarrow \int_{v_0}^0 dv = -\frac{2B}{m\lambda(\sqrt{2} + 1)} \int_0^x x dx$$

$$\Rightarrow x_{\text{max}} = \frac{\sqrt{m\lambda v_0(\sqrt{2} + 1)}}{B}$$



5. A, D

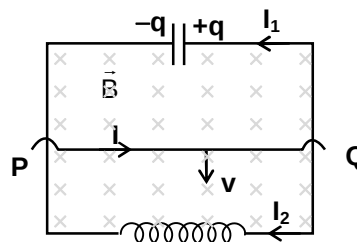
 Sol. $B\ell v = \frac{q}{C}$

$$I_1 = \frac{dq}{dt} = B\ell ca$$

$$B\ell v = L \frac{dI_2}{dt}$$

$$\Rightarrow I_2 = \frac{B\ell x}{L}$$

$$I = I_1 + I_2 = B\ell Ca + \frac{B\ell x}{L}$$



Magnetic force on the wire PQ

$$F_m = I\ell B = B^2 \ell^2 C a + \frac{B^2 \ell^2 x}{L}$$

$$mg - F_m = ma \Rightarrow mg - \left(B^2 \ell^2 C a + \frac{B^2 \ell^2 x}{L} \right) = ma$$

$$\Rightarrow (m + B^2 \ell^2 C) a = mg - \frac{B^2 \ell^2 x}{L}$$

$$\text{when } a = 0, x = \frac{mgL}{B^2 \ell^2}$$

6. B, C

Sol. $\sin \theta = \frac{\ell}{mv_0 / qB_0} = \frac{1}{2}$

$$\theta = 30^\circ$$

Time interval of its motion in magnetic field

$$t_1 = \frac{\theta}{\omega} = \frac{\pi/6}{qB_0/m} = \frac{\pi m}{6qB_0}$$

Time interval of its motion in electric field

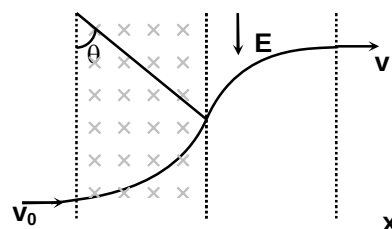
$$t_2 = \frac{v_0 \sin 30^\circ}{qE_0/m} = \frac{mv_0}{2qE_0}$$

$$\text{Further, } \ell = \frac{v_0^2 \sin 60^\circ}{2qE_0/m}$$

$$\Rightarrow \frac{mv_0}{2qB_0} = \frac{mv_0^2 \sin 60^\circ}{2qE_0} \Rightarrow E_0 = \frac{\sqrt{3}v_0 B_0}{2}$$

$$\Rightarrow t_2 = \frac{mv_0}{2q} \frac{2}{\sqrt{3}v_0 B_0} = \frac{m}{\sqrt{3}qB_0}$$

$$\text{Total time } t = t_1 + t_2 = \frac{m}{6qB_0} (\pi + 2\sqrt{3})$$



7. B, C

Sol. Let at any point A of trajectory of ray the tangent to the path of ray makes an angle θ with x-axis. From Snell's law

$$1 \times \sin 90^\circ = n \sin \theta$$

$$\Rightarrow \sin \theta = \frac{1}{n} = \frac{24-x}{24}$$

$$\tan \theta = \frac{24-x}{\sqrt{48x-x^2}}$$

$$\frac{dy}{dx} = \frac{24-x}{\sqrt{48x-x^2}}$$

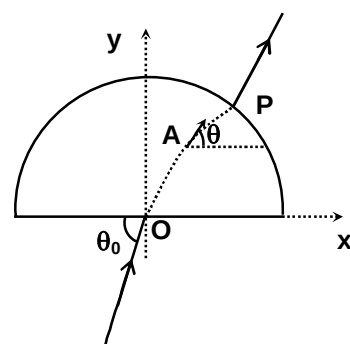
$$\int_0^y dy = \int_0^x \frac{(24-x)dx}{\sqrt{48x-x^2}}$$

$$\Rightarrow x^2 + y^2 - 48x = 0, \text{ so the path of the ray is circular.}$$

$$\text{At P, } x^2 + y^2 = R^2 = 144$$

$$\Rightarrow 48x = 144$$

$$\Rightarrow x = 3 \text{ cm}$$



8. A, C, D

Sol. $r_1 = \frac{2m\ell}{3m} = \frac{2\ell}{3}$

$$r_2 = \ell - r_1 = \frac{\ell}{3}$$

 For motion of star of mass m about centre of mass

$$m\omega^2 r_1 = \frac{2Gm^2}{\ell^2} \Rightarrow \omega = \sqrt{\frac{3Gm}{\ell^3}}$$

Ratio of their kinetic energy about centre of mass frame

$$\frac{k_1}{k_2} = \frac{\frac{1}{2}m\left(\omega\frac{2\ell}{3}\right)^2}{\frac{1}{2}2m\left(\omega\frac{\ell}{3}\right)^2} = 2:1$$

Ratio of their angular momenta w.r.t centre of mass frame

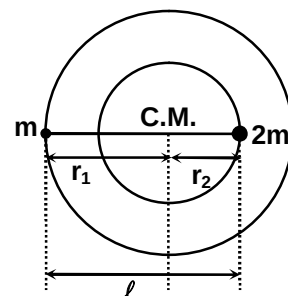
$$\frac{L_1}{L_2} = \frac{m\left(\frac{2\ell}{3}\right)^2 \omega}{2m\left(\frac{\ell}{3}\right)^2 \omega} = 2:1$$

Total angular momentum of the system about centre of mass frame

$$= m\left(\frac{2\ell}{3}\right)^2 \omega + 2m\left(\frac{\ell}{3}\right)^2 \omega = \frac{2}{3}m\omega\ell^2$$

Total kinetic energy of the system w.r.t. centre of mass frame

$$= \frac{1}{2}m\left(\omega\frac{2\ell}{3}\right)^2 + \frac{1}{2}2m\left(\omega\frac{\ell}{3}\right)^2 = \frac{Gm^2}{\ell}$$



9. A, B

Sol. Buoyant force on the rod

$$F_B = V2\rho g = 2mg$$

$$a_{cm} = \frac{\Sigma F_{ext}}{2m} = \frac{2mg - 2mg}{2m} = 0$$

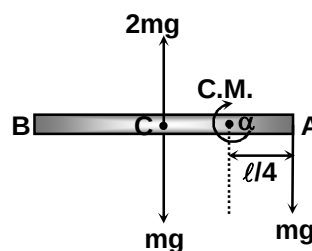
$$I_{cm} = \frac{m\ell^2}{12} + m\left(\frac{\ell}{4}\right)^2 + m\left(\frac{\ell}{4}\right)^2 = \frac{5m\ell^2}{24}$$

$$\tau_{cm} = I_{cm}\alpha \Rightarrow \alpha = \frac{2mg\frac{\ell}{4}}{\frac{5m\ell^2}{24}} = \frac{12g}{5\ell}$$

$$a_A = \alpha r = \alpha\frac{\ell}{4} = \frac{3g}{5} \downarrow$$

$$a_B = \alpha r = \alpha\frac{3\ell}{4} = \frac{9g}{5} \uparrow$$

$$a_C = \alpha r = \alpha\frac{\ell}{4} = \frac{3g}{5} \uparrow$$



10. B, C, D

Sol. Buoyant force on the cone = $v\rho g$

$$= \frac{1}{3}\pi R^2 2R\rho g = \frac{2}{3}\rho g\pi R^3$$

If F_y be the vertical component of the force due to fluid pressure,

$$\rho g 2R \times \pi R^2 - F_y = \frac{2}{3}\rho g\pi R^3$$

$$F_y = \frac{4}{3}\rho g\pi R^3$$

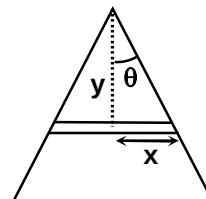
$$T + \frac{2}{3}\rho g\pi R^3 = \frac{1}{3}\pi R^2 2R \times 2\rho g$$

$$T = \frac{2}{3}\rho g\pi R^3$$

Horizontal force exerted by the fluid on the curved surface ABC = Horizontal force exerted by imaginary vertical surface AB by the fluid.

$$F_x = \int dF_x = \int_0^{2R} (\rho g y) 2x dy$$

$$= \int_0^{2R} \rho g y^2 dy = \frac{8\rho g R^3}{3}$$

**Section – B**

11. 4

Sol. Electric field inside the cavity is uniform and given by $\vec{E} = \frac{\rho \vec{r}}{2\epsilon_0}$ where $\vec{r}$ is the vector joining point

O to point O'

$$\text{Range } R = \frac{v_0^2 \sin 2\theta}{qE/m}$$

$$\Rightarrow R = \frac{mv_0^2 \sin 2\theta}{q\rho \frac{R}{2\epsilon_0}}$$

$$\Rightarrow v_0 = \sqrt{\frac{q\rho R^2}{4\epsilon_0 m \sin 2\theta}}$$

$$v_{\min} = \sqrt{\frac{q\rho R^2}{4\epsilon_0 m}}$$

12. 6

 Sol. Let at certain moment its direction of motion makes an angle θ with the vertical as shown.

$$5 \cos \theta - 5 = ma_t$$

$$5 - 5 \cos \theta = ma_y$$

$$\Rightarrow a_t = -a_y$$

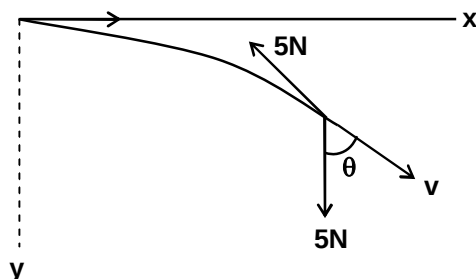
$$\Rightarrow \frac{dv}{dt} = -\frac{dv_y}{dt} \Rightarrow v = -v_y + c$$

$$\text{When } v = 9 \text{ m/s, } v_y = 0$$

$$\Rightarrow c = 9$$

$$v = -v \cos \theta + 9$$

$$\Rightarrow v = \frac{9}{1 + \cos \theta} = \frac{9}{1 + \cos 60^\circ} = 6 \text{ m/s}$$



13. 4

Sol. When the block leaves the surface

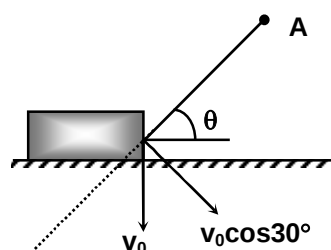
$$T \sin 30^\circ = mg$$

At the moment the block leaves surface it moves in circular path about the end A of the string with velocity $v_0 \cos 30^\circ$

$$\Rightarrow T - mg \sin 30^\circ = \frac{m}{\ell} (v_0 \cos 30^\circ)^2$$

$$\Rightarrow \frac{mg}{\sin 30^\circ} - mg \sin 30^\circ = \frac{m}{\ell} v_0^2 \cos^2 30^\circ$$

$$\Rightarrow v_0 = \sqrt{\frac{g\ell}{\sin 30^\circ}} = 4 \text{ m/s}$$



Section - C

14. 02.12

15. 00.36

Sol. (for Q. 14 to 15):

$$\text{The energy of photon incident on plate A} = \frac{1240 \text{ eV} \cdot \text{nm}}{500 \text{ nm}} = 2.48 \text{ eV}$$

The maximum kinetic energy of emitted photoelectrons from plate A

$$K_{\max} = h\nu - \phi = (2.48 - \phi) \text{ eV}$$

The maximum kinetic energy of photoelectrons reaching the plate B = $k_{\max} + e \times 1.64$

$$= (2.48 - \phi + 1.64) \text{ eV}$$

The maximum energy of photons emitted from plate B

$$= \frac{1240 \text{ eV} \cdot \text{nm}}{620 \text{ nm}} = 2 \text{ eV}$$

The maximum energy of emitted photons from plate B = maximum kinetic energy of photoelectrons striking the plate B

$$\Rightarrow 2 \text{ eV} = (2.48 - \phi + 1.64) \text{ eV} \Rightarrow \phi = 2.12 \text{ eV}$$

16. 00.50

17. 00.25

Sol. (for Q. 16 to 17):

$$\text{Path difference at O} = d \sin \theta_0 - (\mu - 1)t$$

$$= 5 \times 10^{-4} \times \frac{1}{50} - (1.5 - 1) \times 19.75 \times 10^{-6} = 1.25 \times 10^{-7}$$

Phase difference at O

$$\phi = \frac{2\pi}{\lambda} \times \Delta x = \frac{2\pi}{5 \times 10^{-7}} \times 1.25 \times 10^{-7} = \frac{\pi}{2}$$

$$I = I_0 \cos^2\left(\frac{\phi}{2}\right) = I_0 \cos^2\left(\frac{\pi}{4}\right)$$

$$\frac{I}{I_0} = 0.50$$

$$-d \sin \theta_0 + d \frac{y}{D} + (\mu - 1)t = n\lambda$$

$$\Rightarrow +0.125 \times 10^{-6} + \frac{0.5 \times 10^{-3}}{1} y = n\lambda$$

$$\Rightarrow y = (n \times 5 \times 10^{-7} - 1.25 \times 10^{-7}) 2 \times 10^3$$

$$y_{\min} = 2.5 \times 10^{-4} \text{ m} = 0.25 \text{ mm}$$

18. 5400.00

19. 4600.00

Sol. (for Q.18 to 19):

$$\lambda_m T_B = b \Rightarrow T_B = \frac{2.9 \times 10^{-3}}{7.25 \times 10^{-7}} = 4000 \text{ K}$$

$$\frac{dQ}{dt} = -kA \frac{dT}{dx}$$

$$\Rightarrow 33.6 = -k_0(1 + 0.2x)A \frac{dT}{dx}$$

$$\Rightarrow \frac{33.6}{k_0 A} \int_0^5 \frac{dx}{1 + 0.2x} = - \int_{T_A}^{T_B} dT$$

$$\Rightarrow \frac{33.6}{8.4 \times 10^{-2} \times 0.2} \ln 2 = (T_A - T_B)$$

$$\Rightarrow \frac{33.6}{8.4 \times 10^{-2} \times 0.2} \times 0.7 = T_A - 4000$$

$$\Rightarrow T_A = 5400 \text{ K}$$

If temperature at mid point of rod is T_C

$$\Rightarrow \frac{33.6}{k_0 A} \int_0^{2.5} \frac{dx}{1 + 0.2x} = - \int_{T_A}^{T_C} dT$$

$$\Rightarrow T_C = 4600 \text{ K}$$

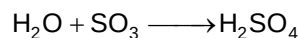
Chemistry

PART – II

Section – A

20. C

Sol. Let the maximum limiting labelling of oleum is $(100 + x)\%$
 In 100 gram of oleum the maximum mass of free SO_3 should be tending to 100 gram and hence, the mass of water needed x gram, should be exactly that mass which combine completely with all the free SO_3 present

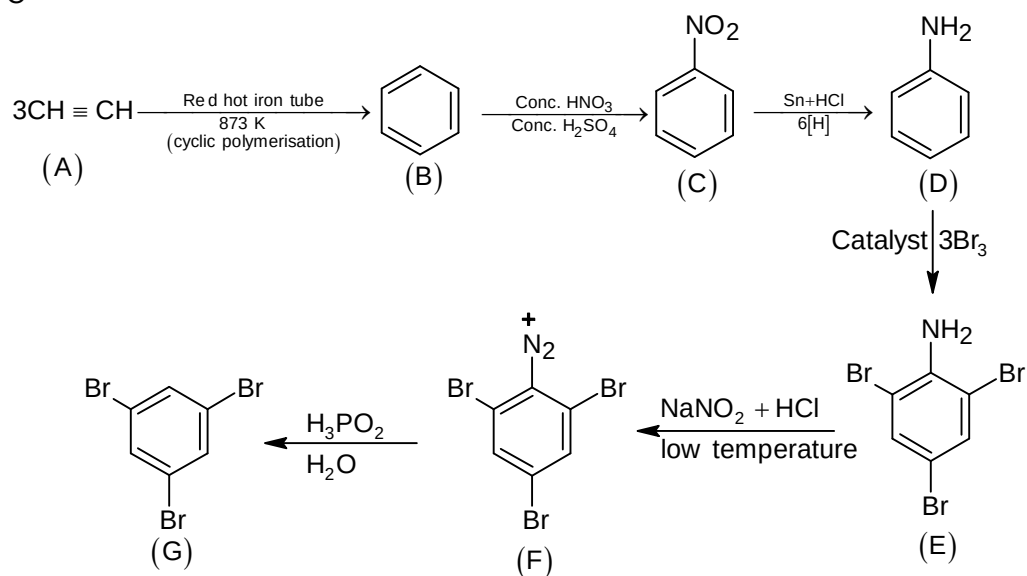


$$x = \frac{100}{80} \times 18$$

$$= 22.5$$

21. C

Sol.



22. C

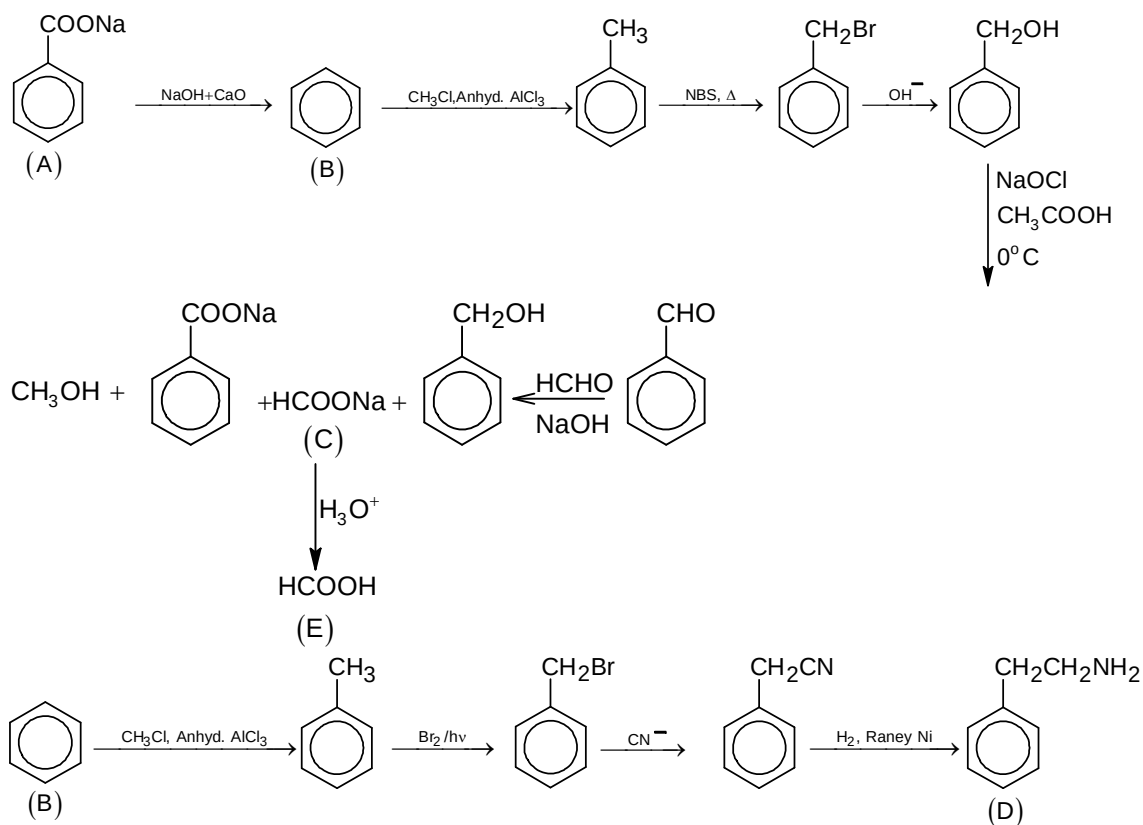
23. B

Sol. As $P_{\text{ext}} = 0$ $W = -P_{\text{ex}} \Delta V = 0$
 As $\Delta T = 0$ $\Delta E = nC_V \Delta T = 0$
 But $\Delta S = nR \ln \frac{V_2}{V_1}$

24. B, D

25. A, B, D

Sol.



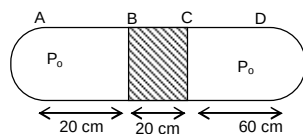
26. A, B, C, D

27. A, B, C, D

28. A, B, C, D

29. A, D

Sol.



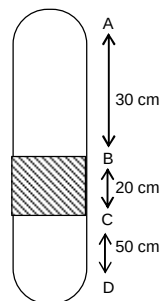
For column AB

$$P_1 V_1 = P_2 V_2$$

$$P_0 \times 20A = P_2 \times 30A$$

$$P_2 = \frac{P_0 \times 20A}{30A} = \frac{2P_0}{3}$$

For column CD



$$P_1 V_1 = P_2' V_2$$

$$P_0 \times 60A = P_2 \times 50A$$

$$P_2' = \frac{6P_0}{5}$$

$$P_2' = P_2 + 20$$

$$\frac{6P_0}{5} = \frac{2P_0}{3} + 20$$

$$P_0 = 37.5 \text{ cm Hg}$$

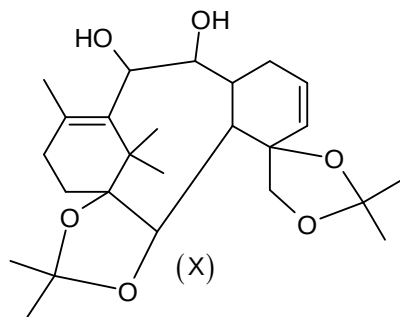
Section – B

30. 5

Sol. Pb^{2+} , Bi^{3+} , Cd^{2+} , Sb^{3+} , Mn^{2+}

31. 35

Sol.



$$a = 7$$

$$b = \frac{7 \times 10}{2} = 35$$

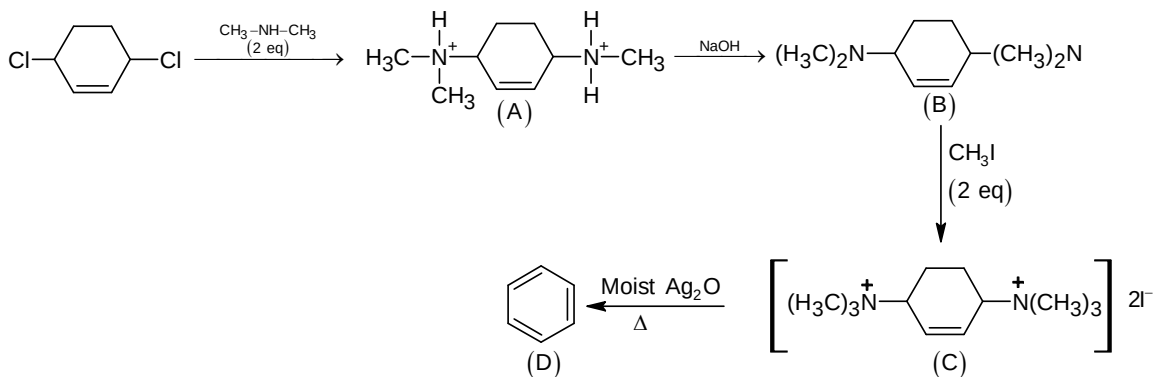
32. 7

Sol. Methylene blue and TiO_2 are positively charged sol.

Section – C

33. 156.00

Sol.



$$x = \frac{\text{M. wt. of } \text{C}_6\text{H}_6}{0.5} = \frac{78}{0.5} = 156$$

34. 04.00

Sol.

$$y = \frac{(2 \times 6 + 2) - 6}{2}$$

$$= \frac{14 - 6}{2} = \frac{8}{2} = 4$$

35. 02.00

Sol. $x = 1 \times 2 = 2$

Soil pollutants = DDT

Water pollutant = Pb, $\text{Cl}_2\text{C} = \text{CCl}_2$

36. 00.50

Sol. $y = \frac{1}{2} = 0.5$

Stratospheric pollutants = 1

37. 00.60

$$\begin{aligned} \text{Sol. } x &= \frac{2^n - 2^{\frac{n}{2}-1}}{100} \\ &= \frac{2^6 - 2^{\frac{6}{2}-1}}{100} \\ &= \frac{64 - 4}{100} = \frac{60}{100} = 0.6 \end{aligned}$$

38. 1000.00

$$\begin{aligned} \text{Sol. } &\frac{2^{n-1} + 2^{\frac{n}{2}-1}}{0.01} \\ &= \frac{2^{4-1} + 2^{\frac{4}{2}-1}}{0.01} \\ &= \frac{8 + 2}{0.01} = \frac{10}{0.01} = 1000 \end{aligned}$$

Mathematics

PART – III

Section – A

39. D

Sol. As per Fermet's Theorem, for co-prime numbers a and p ,
 Remainder when a^p is divided by $p = a$
 $\Rightarrow$ Remainder when 2^{199} is divided by $199 = 2$
 $\Rightarrow$ Remainder when 2^{1990} is divided by $199 = 29$
 $\Rightarrow$ Remainder when 2^{1990} is divided by $10 = 4$
 $\Rightarrow$ Remainder $N_1 = 199\lambda_1 + 29 = 10\lambda_2 + 4$
 $\Rightarrow$ Least $N_1 = 1024$

40. A

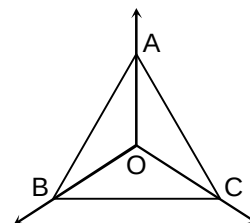
Sol. Let the proposes of the problem be called x and the friends be denoted as A, B, C, D, E, F. Since x dines with all the 6 friend exactly on one day. So we have combination XABCDEF for one day. Next combination is XABCDEF, XABCDF, ${}^6C_5 = 6$
 So, for x has dined with every four friends 3 times with just one of them he has dined
 So, for 6 days. Hence he has to dine with every one of them for one more day
 Thus, total number of days he dined so for with atleast one of his friends $1 + 6 + 6 = 13$ days

41. B

Sol. Set $\alpha = \tan^{-1}\left(\frac{1}{5}\right)$; $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{5}{12}$; $\tan 4\theta = \frac{2\tan 2\theta}{1-\tan^2(2\theta)} = \frac{120}{119}$
 $\therefore \tan\left(4\theta - \frac{\pi}{4}\right) = \frac{1}{239}$

42. B

Sol. $x + 2y + 3z \leq 6$ we need to find volume of tetrahedron
 $v = \left(\frac{1}{6} \times 6 \times 3 \times 2\right) \times 8$



43. A, D

Sol. If $\lambda < 0$, then sum $= \sum_{k=0}^{n-1} \frac{\pi + 2k\pi}{n} = n\pi$
 If $\lambda > 0$, then sum $= \sum_{k=0}^{n-1} \frac{2k\pi}{n} = (n-1)\pi$

44. A, B, D

Sol. Here, $P(1) = 0$, $P(2) = \frac{1}{3}$, $P(n) = (1 - P(n-1))\frac{1}{3}$, $n \geq 3$

45. A, B, C

Sol. Take $f(x) = x^2 + 2ax + 3bx^2$

46. A, D

Sol. $f(x)$ is discontinuous when $\cos x = \frac{-1+\sqrt{5}}{2}$

47. A, B, C

Sol. Here, $CP \perp CQ$

48. A, D

Sol. Here, $x \in [-\sqrt{3}, \sqrt{3}]$ and $y = \frac{1}{\sqrt{3}}$ **Section – B**

49. 4

Sol. Here, $S = \sum_{k=1}^{624} (k+1)^{\frac{1}{4}} - k^{\frac{1}{4}}$

50. 6

Sol. Here, $k_0 = 1$, $k_1 = 4\lambda$, $k_2 = 6\lambda^2 + 4\mu$
Also, $k_0 = k_1 = k_2$

51. 11

Sol. The locus of the centre of the variable circle S' is a parabola with directrix as $x + 2 = 0$ and focus at $(4, 0)$ **Section – C**

52. 04.00

53. 11.00

Sol. (for Q. 52 to 53):

$$\sum_{r=0}^{p_2} (p_1 r - 3) \cdot {}^{p_2}C_r = (p_1 p_2 - 6) p^{p_2-1}$$

54. -00.25

55. 00.75

Sol. (for Q. 54 to 55):

$$\text{Here, } f'(x) = 2px + q \Rightarrow f(x) = -\frac{x^2}{4} - \frac{x}{2} + 1$$

56. 08.00

57. 10.00

Sol. (for Q. 56 to 57):

$$\text{Here, } f(x) = \frac{\left(x - \frac{1}{x}\right)}{\left(x - \frac{1}{x}\right)\left(\left(x - \frac{1}{x}\right)^2 + 3\right) + 2}$$